

Flight Route Requests

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities with airports but no flight connections. You are given m requests which routes should be possible to travel.

Your task is to determine the minimum number of one-way flight connections which makes it possible to fulfil all requests.

Input

The first input line has two integers n and m : the number of cities and requests. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines describing the requests. Each line has two integers a and b : there has to be a route from city a to city b . Each request is unique.

Output

Print one integer: the minimum number of flight connections.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
4 5 1 2 2 3 2 4 3 1 3 4	4

Explanation: You can create the connections $1 \rightarrow 2$, $2 \rightarrow 3$, $2 \rightarrow 4$ and $3 \rightarrow 1$. Then you can also fly from city 3 to city 4 using the route $3 \rightarrow 1 \rightarrow 2 \rightarrow 4$.